Prime-Leaf Tree Theory

Arithmetic as the statistical mechanics of multiplication histories

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Abstract

We treat the integers as a gas with one mode per prime p, of energy $\log p$ (the setting of Julia's gas of numbers and of Bost–Connes), and study it before symmetrization and before the interaction history is forgotten. A microstate is a multiplication tree: an ordered, parenthesized record of pairwise combinations whose only conserved quantity is the integer it evaluates to. Forgetting history and symmetrizing are independent coarse-grainings, giving four gases whose partition functions are exact functionals of the prime zeta function $P(\beta)$; their inverse limiting temperatures form a strict ladder $1 < 1.3994\ldots < 1.9885\ldots < \sigma^* = 2.5974\ldots$, and whether the limiting temperature is attained is decided by associativity: the symmetrized gases have poles, the tree gases square-root branch points, with $Z_{\mathcal{T}}(\sigma^*) = \frac{1}{2}$ exactly. The complex-temperature zeros of the tree gas inherit the Riemann zeros through P; their new branch points, the solutions of $P(\beta) = \frac{1}{4}$, do not lie on a vertical line but carry an exact asymptotic density given by mode-dominance probabilities (total density $0.1436\ldots$ in $\operatorname{Re}\beta > 1$), and a census of 595 zeros to height 5000 reveals a spectrum far more rigid than Poisson or random-matrix statistics. The Fundamental Theorem of Arithmetic appears as zero residual entropy of the coarse-grained description, and the classical laws of arithmetic hold only in expectation. All results are reproducible from an accompanying open-source package.

Keywords: prime numbers; partition functions; limiting (Hagedorn) temperature; complex-temperature (Fisher) zeros; almost periodic functions; statistical bootstrap; value distribution

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1 Introduction

1.1 Multiplication before evaluation

Ordinary arithmetic writes

$$3 \times 2 = 6 \qquad \text{and} \qquad 1+1+1+1+1+1 = 6$$

and treats the two results as the same kind of object. The starting intuition of this paper is that they are not, and one liter of water is enough to see it. Pour six one-liter bottles into a tank: that is addition, and it produces a flat six liters. Now take 3×2 liters seriously as a construction: three cartons, each holding two bottles. Take 2×3 liters: two cartons, each holding three. Both measure six liters; neither *is* six liters; and they are not each other. The slogan:

You never get six liters from multiplication. You get a structured object whose shadow is 6*.*

Physics has standard names for this situation: 6 is a *macrostate*, and the packings that produce it are *microstates*. We therefore refuse to let multiplication collapse into numbers. A product is a *multiplication tree*: the recorded history of how the primes involved were combined, ordered and parenthesized, unevaluated. (The cartons and bottles return in §6, where the slogan becomes a theorem.) The familiar integer is produced only by an evaluator, the shadow map N, which plays the role of the measurement: it reads off the single conserved quantity and discards the history. The classical laws of multiplication are then properties of the coarse-graining rather than of the multiplication itself: they record which distinct microstates the measurement fails to distinguish. In one line:

The primes are the particles; an integer is a measured energy; the Riemann zeta function is a partition function; and ordinary arithmetic is what survives total coarse-graining.

1.2 The dictionary

The paper is written for readers who think in statistical mechanics. Every physical term below is either an *exact structural identity* (the object on the left satisfies, verbatim, the defining equation of the object on the right) or is flagged as imagery.

tree theory	physics
prime p	a mode of energy $\varepsilon_p = \log p$
multiplication tree $T \in \mathcal{T}$	a composite: a recorded history of pairwise combinations
$\boxtimes$ (pair formation)	combination; on energies $E = \log N$ it is addition
shadow $N(T)$	the measurement: the measured integer $e^{E(T)}$
left–right order kept	ordered channels, as for planar diagrams
parenthesization kept	interaction history retained
quotient by associativity	forget the history, keep the particle list: asymptotic state
quotient by commutativity	Bose symmetrization
words $\mathcal{W}$	basis of the full (Boltzmann) Fock space
multisets $\mathcal{M}$	occupation numbers: the gas of numbers of Julia
degeneracy $g(n)$; $S(n) = \log g(n)$	microstate count and Boltzmann entropy of the integer n
Fund. Thm. of Arithmetic	the occupation-number description is faithful: $S_{\mathcal{M}} \equiv 0$
$Z(s) = \sum N^{-s}$, $s = \beta$	canonical partition function at inverse temperature β
$Z = P + Z^2$	self-consistency equation (the Hagedorn–Frautschi bootstrap)
critical exponents σ	inverse limiting ("Hagedorn") temperatures β_H
solutions of $P(s) = \frac{1}{4}$	complex-temperature zeros of Z (its "Fisher zeros")
grouped quantities $\mathcal{G}$	states of hierarchically clustered matter
the action $\rho(T)$	a process applied to states; $\boxtimes \mapsto$ composition
flattening α	the extensive observable (particle number); expectation level

Table 1: The dictionary. The "measurement" and "clustered matter" rows are imagery; the limiting temperature and complex-temperature zeros carry the standard names (Hagedorn, Fisher) where the corresponding work is cited in the text. Every other row is an exact identity.

Everything is elementary and self-contained: finite combinatorics plus one-variable complex analysis. No result depends on the imagery.

1.3 Results

1. **The microstate space** (§2). The world of trees is the free binary-tree structure $\mathcal{T}$ on the set of primes; the measurement is the unique multiplication-preserving map $N\colon \mathcal{T} \to (\mathbb{N}_{\geq 2}, \cdot)$ with $N(L_p) = p$; energies $E = \log N$ add under combination. The interacting world has no vacuum: 1 is not the shadow of any tree, and the empty state exists only in the Fock description.

2. **Coarse-graining is two-dimensional** (§3). Forgetting the interaction history (associativity) and symmetrizing (commutativity) are *independent* operations, giving a commuting square of four state spaces $\mathcal{T}, \mathcal{W}, \mathcal{B}, \mathcal{M}$. The Fundamental Theorem of Arithmetic says exactly that the final corner has one microstate per macrostate. Each level carries a computable entropy: e.g. $S_{\mathcal{T}}(12) = \log 6$. The degeneracies are Catalan $\times$ multinomial for $\mathcal{T}$, multinomial for $\mathcal{W}$ (OEIS A008480), a sequence for $\mathcal{B}$ that restricts to the Wedderburn–Etherington numbers on prime powers (OEIS A375120, whose formula and program were corrected following this work), and 1 for $\mathcal{M}$. All verified exhaustively for $n \leq 20000$.

3. **Four partition functions and a temperature ladder** (§4). All four are exact functionals of the prime zeta function P: the free Bose gas $\zeta(\beta)$ (the Euler product is the textbook free-boson computation), a geometric gas $P/(1-P)$, an Otter gas with argument-doubling equation, and the planar tree gas obeying the self-consistency equation $Z = P + Z^2$. The inverse limiting temperatures $\beta_H = 1/T_H$ form a strict ladder

$$1 \;<\; 1.3994333287263303 \;<\; 1.9884768518009081 \;<\; \sigma^* = 2.5973851271346716,$$

one rung per layer of remembered structure. At its limiting temperature the tree gas converges with $Z_{\mathcal{T}}(\sigma^*) = \frac{1}{2}$ exactly and a square-root branch point: the universal branched-polymer singularity, the same criticality that governs planar-diagram counting in the Gaussian matrix model. The symmetrized gases instead have poles: whether the limiting temperature is attainable is decided by associativity. The classical Hagedorn–Frautschi–Nahm bootstrap, transplanted onto the prime spectrum, slots into the same family with $\beta_H = 2.1487511659\ldots$ (Boltzmann counting) and $2.3072314551\ldots$ (Bose counting).

4. **Complex-temperature zeros** (§5). $Z_{\mathcal{T}}$ is algebraic in P, so its complex-temperature singularities are inherited from P (which carries a branch point at $\beta = \rho/k$ for every nontrivial Riemann zero ρ, and has the imaginary axis as a natural boundary), plus new branch points where $P(\beta) = \frac{1}{4}$. We compute the five with $0 < \operatorname{Im}\beta < 50$: they recur near integer multiples of $2\pi/\log 2$ (the lightest mode, of energy $\log 2$, sets the frequency) and do not lie on a vertical line. The naive tree analogue of the Riemann Hypothesis is therefore false, and the right question concerns the distribution of these zeros: in every substrip of $\operatorname{Re} s > 1$ the tree zeros have an exact asymptotic density (Theorem 5.3, proved via the Jessen–Tornehave theory of almost periodic functions), with total density $D(1) = 0.1436\ldots$ in the strip $\operatorname{Re} s > 1$ (Corollary 5.7). Each zero occurs when a single mode outshines the coherent rest, and the density is the total dominance rate $\frac{1}{2\pi}\sum_p (\log p)\, q_p$. A census of 595 zeros to height 5000 confirms the density and reveals a spectrum markedly more rigid than either Poisson or random-matrix statistics (§5.4), the signature of the zeros of an almost periodic function.

5. **The additive bridge** (§6). Trees act as *processes* on states of clustered matter (boxed heaps of unit quanta), and the classical laws stratify by observability: associativity is

invisible to the action (exactly, because processes compose), while commutativity fails on states and holds only in expectation, and distributivity holds only in expectation. "You never get six liters from multiplication" becomes a theorem about the image of the action. The primes themselves are then derivable inside the additive world: a size is composite iff its heap can crystallize into $k \geq 2$ identical cells of $m \geq 2$ quanta; primes are the non-crystallizable sizes.

6. **Computational content** (§7). Microstate counting is *shape-blind*: every degeneracy depends only on the exponent profile of n, so no counting of histories can locate primes. In the other direction, a random integer seen at full resolution is a universal random object: a uniform Catalan tree with a Gaussian, $\log \log x$, number of prime factors (Erdős–Kac lifted to trees). The interface where the theory genuinely meets the complexity of finding primes is *expression cost*: integer complexity is the first invariant here that escapes P (it distinguishes 4 from 9), Wilson's theorem makes primality one factorial-expression away, and the open frontier, deterministic prime generation, is a derandomization problem, which the theory maps but does not shorten.

1.4 Relation to existing work

The Fock-level reading is classical: the gas of numbers of Julia [23] and Spector [36], deepened by Bost–Connes [5], identifies $\zeta(\beta)$ with a free bosonic partition function over the spectrum $\{\log p\}$ and locates a limiting temperature at $\beta = 1$. The statistical bootstrap and its limiting temperature are from [21, 19], with the analytical solution and the square-root criticality due to Nahm [31]. Lee–Yang and Fisher zeros are from [27, 16]. On the mathematical side, the free binary-tree structure on a set and its Catalan combinatorics are standard [37]; the degeneracy $g_{\mathcal{T}}(n)$ appears as OEIS A144757 and the ordered-factorization count as A008480 [35]; ordered factorizations with arbitrary parts go back to Kalmár [25]; the natural boundary of the prime zeta function is due to Landau and Walfisz [26]. Closest in spirit to the unsymmetrized viewpoint is Loday's *arithmetree* [28, 6]: arithmetic on planar binary trees with non-commutative addition and one-sided distributivity.

Against this background the contributions here are: the two-dimensional coarse-graining (the square, with the new non-planar corner, its Otter equation and its limiting temperature); the exact critical value $Z_{\mathcal{T}}(\sigma^*) = \frac{1}{2}$ and the attainability dichotomy; the computation and non-collinearity of the complex-temperature zeros; the process/state/expectation stratification of the arithmetic laws; and the internal derivation of the primes. One caution runs through the paper: a free structure by itself carries no information about the primes. Everything classically hard enters through the single-particle spectrum, that is through $P(\beta)$, or through the additive bridge. What the theory offers is a clean separation of the structural from the arithmetic, and computable invariants in the separated layers.

2 Multiplication trees

2.1 Modes and combination histories

Definition 2.1 (Multiplication trees)**.** Fix the set of prime leaves $\{L_p : p \text{ prime}\}$. The set $\mathcal{T}$ of *multiplication trees* is defined by the grammar

$$T \;::=\; L_p \;\mid\; (T_1 \boxtimes T_2).$$

Thus $\mathcal{T}$ is the *free binary-tree structure* on the set of primes: $\boxtimes$ is ordered-pair formation, subject to no laws whatsoever.

Physical reading. L_p is one particle in the mode of energy $\varepsilon_p = \log p$. A tree is a composite that remembers how it was assembled: $T_1 \boxtimes T_2$ is the combination of T_1 and T_2, in that order. Examples of *distinct* microstates:

$$L_2, \qquad L_2 \boxtimes L_3, \qquad L_3 \boxtimes L_2, \qquad (L_2 \boxtimes L_3) \boxtimes L_5, \qquad L_2 \boxtimes (L_3 \boxtimes L_5).$$

Non-commutativity and non-associativity are not axioms we impose; they are the absence of axioms. Keeping the left–right order is keeping the combination channels ordered, as one does for planar diagrams of matrix-valued fields [39]; keeping the parenthesization is keeping the interaction history: which pair combined first. A multiplication tree is the parse tree of a multiplication expression before evaluation; equivalently, a tree-level diagram built from a single cubic vertex whose external legs are labeled by primes.

For $T \in \mathcal{T}$ write $|T|$ for the number of leaves and $\mathrm{word}(T)$ for the left-to-right sequence of leaf labels; $\mathrm{supp}(T)$ denotes the set of primes occurring.

2.2 The measurement

Definition 2.2 (Shadow)**.** The *shadow map* $N\colon \mathcal{T} \to \mathbb{N}_{\geq 2}$ is defined recursively by

$$N(L_p) = p, \qquad N(T_1 \boxtimes T_2) = N(T_1)\, N(T_2).$$

Equivalently: N is the unique multiplication-preserving map from $\mathcal{T}$ to the multiplicative semigroup $(\mathbb{N}_{\geq 2}, \cdot)$ sending L_p to p (that is, the unique map with $N(L_p) = p$ that respects $\boxtimes$).

Physical reading. Define the energy $E(T) = \log N(T)$. Then $E(L_p) = \varepsilon_p$ and $E(T_1 \boxtimes T_2) = E(T_1) + E(T_2)$: energies add under combination *because* shadows multiply. The shadow is the measured value: the single conserved quantity, blind to history and to order. An integer is the reading e^E shared by many microstates, and is not itself a microstate.

Remark 2.3 (No vacuum)*.* $\mathcal{T}$ has no unit object and 1 is not the shadow of any tree: the image of N is exactly $\mathbb{N}_{\geq 2}$. One could freely adjoin an empty tree I, but then $I \boxtimes T \neq T$ (freeness again), which buys nothing. In gas language: the interacting world has no vacuum state; the vacuum ($n = 1$, zero energy) exists only in the Fock description, as the empty occupation configuration (§3).

Composite numbers are not primitive objects of this world. There is no L_6; the value 6 occurs only as the common reading of the two trees $L_2 \boxtimes L_3$ and $L_3 \boxtimes L_2$. Likewise 8 is the common reading of $(L_2 \boxtimes L_2) \boxtimes L_2$ and $L_2 \boxtimes (L_2 \boxtimes L_2)$: distinct formation histories, one observable. Quantifying this degeneracy is the subject of the next section.

3 Coarse-graining: from histories to Fock space

3.1 Two independent coarse-grainings

Definition 3.1. Let $\approx_a$ be the smallest congruence on $(\mathcal{T}, \boxtimes)$ with $(T_1 \boxtimes T_2) \boxtimes T_3 \approx_a T_1 \boxtimes (T_2 \boxtimes T_3)$ for all T_i, and $\approx_c$ the smallest congruence with $T_1 \boxtimes T_2 \approx_c T_2 \boxtimes T_1$. Write

$$\mathcal{W} = \mathcal{T}/{\approx_a}, \qquad \mathcal{B} = \mathcal{T}/{\approx_c}, \qquad \mathcal{M} = \mathcal{T}/({\approx_a} \vee {\approx_c}).$$

$$\begin{tikzcd}
\boxed{\begin{array}{c} \mathcal{T} \quad \text{planar trees} \quad g_{\mathcal{T}}(12) = 6 \\ (L_2L_2)L_3,\ (L_2L_3)L_2,\ (L_3L_2)L_2, \\ L_2(L_2L_3),\ L_2(L_3L_2),\ L_3(L_2L_2) \end{array}} \arrow[r, "\text{forget history}"] \arrow[d, "\text{symmetrize}"'] & \boxed{\begin{array}{c} \mathcal{W} \quad \text{words} \quad g_{\mathcal{W}}(12) = 3 \\ (2,2,3), \quad (2,3,2), \quad (3,2,2) \end{array}} \arrow[d, "\text{symmetrize}"] \\
\boxed{\begin{array}{c} \mathcal{B} \quad \text{unordered trees} \quad g_{\mathcal{B}}(12) = 2 \\ \{\{L_2, L_2\}, L_3\}, \quad \{\{L_2, L_3\}, L_2\} \end{array}} \arrow[r, "\text{forget history}"'] & \boxed{\begin{array}{c} \mathcal{M} \quad \text{multisets} \quad g_{\mathcal{M}}(12) = 1 \\ \{2,2,3\} \xrightarrow{N} 12 \end{array}}
\end{tikzcd}$$

Figure 1: The coarse-graining square, shown collapsing the integer $12 = 2^2 \cdot 3$. Forgetting the interaction history (associativity, horizontal) and symmetrizing the particles (commutativity, vertical) are independent, and commute. The number of microstates over 12 drops $6 \to 3 \to 1$ and $6 \to 2 \to 1$ along the two routes; the Fundamental Theorem of Arithmetic is the statement that the final corner has exactly one. (Here L_aL_b abbreviates $L_a \boxtimes L_b$.)

Each quotient is a familiar free object: $\mathcal{W}$ is the free semigroup on the primes (nonempty *words* in primes, i.e. ordered prime factorizations); $\mathcal{B}$ is the free *commutative* binary-tree structure (trees with unordered branches); $\mathcal{M}$ is the free commutative semigroup (nonempty finite *multisets* of primes). The shadow map descends to all four corners, and the two identifications commute (Figure 1). The shadow map of the final corner is where classical arithmetic begins:

Proposition 3.2 (FTA as exactness of the collapse)**.** *The induced map $N \colon \mathcal{M} \to \mathbb{N}_{\geq 2}$ is a bijection. This statement is equivalent to the Fundamental Theorem of Arithmetic.*

Proof. Elements of $\mathcal{M}$ are nonempty prime multisets $\{p_1, \ldots, p_k\}$ with $N = p_1 \cdots p_k$. Surjectivity of N is the existence of a prime factorization; injectivity is its uniqueness. $\square$

Physical reading. The two arrows are the two things an observer can throw away. *Forgetting the history* keeps only the ordered list of constituent particles: an asymptotic state, with the internal dynamics integrated out. *Symmetrizing* declares the particles indistinguishable: Bose statistics. Doing both lands in the occupation-number description: an element of $\mathcal{M}$ is exactly a configuration $(\alpha_2, \alpha_3, \alpha_5, \ldots)$, a basis vector of the symmetric Fock space over the single-particle spectrum. $\mathcal{W}$, the half-way house that keeps order but forgets history, is a basis of the full (Boltzmann) Fock space. Proposition 3.2 then says that the occupation-number description of the integers is faithful: unique factorization is the absence of residual entropy at the Fock level.

Remark 3.3 (The identifications are themselves structured)*.* The associativity identification is not a group action: two trees with the same word are connected by a sequence of reassociations, and the set of parenthesizations of a fixed word carries the *Tamari lattice* structure, realized geometrically by the associahedron [38]. Commutativity moves similarly generate the permutohedron. The quotient square therefore has polytopal fibers, a refinement we do not pursue here.

3.2 Degeneracies: the entropy of an integer

For $n \geq 2$ with factorization $n = \prod_p p^{\alpha_p}$ write $\Omega(n) = \sum_p \alpha_p$. For each corner $\mathcal{X}$ of the square let

$$g_{\mathcal{X}}(n) = \#\{X \in \mathcal{X} : N(X) = n\}, \qquad S_{\mathcal{X}}(n) = \log g_{\mathcal{X}}(n),$$

the microstate count and the Boltzmann entropy of the macrostate n at level $\mathcal{X}$. Proposition 3.2 says $S_{\mathcal{M}} \equiv 0$; the other three levels have genuine entropy.

Proposition 3.4 (Planar trees)**.**

$$g_{\mathcal{T}}(n) \;=\; C_{\Omega(n)-1}\,\frac{\Omega(n)!}{\prod_p \alpha_p!}, \qquad C_k = \frac{1}{k+1}\binom{2k}{k}.$$

Proof. A tree with shadow n has exactly $\Omega(n)$ leaves, by induction and multiplicativity of N. The map $T \mapsto (\mathrm{shape}(T), \mathrm{word}(T))$ is a bijection from $\{T : N(T) = n\}$ to (binary tree shapes with $k = \Omega(n)$ leaves) $\times$ (sequences of primes with product n). There are C_{k-1} shapes [37] and $\Omega!/\prod \alpha_p!$ sequences (permutations of the prime multiset). The two factors are independent: the fiber of the full collapse $\mathcal{T} \to \mathcal{M}$ splits as histories $\times$ orderings. □

Proposition 3.5 (Words)**.** *$g_{\mathcal{W}}(n) = \Omega(n)!/\prod_p \alpha_p!$, with recursion $g_{\mathcal{W}}(n) = \sum_{p|n} g_{\mathcal{W}}(n/p)$ and $g_{\mathcal{W}}(p) = 1$. This is OEIS A008480.*

Proposition 3.6 (Non-planar trees)**.** *$g_{\mathcal{B}}$ satisfies $g_{\mathcal{B}}(p) = 1$ and, for composite n,*

$$g_{\mathcal{B}}(n) \;=\; \frac{1}{2}\sum_{\substack{d|n\\ 1<d<n}} g_{\mathcal{B}}(d)\, g_{\mathcal{B}}(n/d) \;+\; \frac{1}{2}\,[\,n \textit{ is a square}\,]\; g_{\mathcal{B}}(\sqrt{n}).$$

On prime powers, $g_{\mathcal{B}}(p^k)$ is the k-th Wedderburn–Etherington number (OEIS A001190): 1, 1, 1, 2, 3, 6, 11, 23, ...

Proof. An element of $\mathcal{B}$ over composite n is an unordered pair $\{B_1, B_2\}$ with $N(B_1)N(B_2) = n$. Ordered pairs are counted by $\sum_{1<d<n} g_{\mathcal{B}}(d)g_{\mathcal{B}}(n/d)$; the diagonal (possible only when n is a square, both factors of shadow $\sqrt{n}$ and equal as trees) is counted by $g_{\mathcal{B}}(\sqrt{n})$; unordered pairs with repetition are (ordered + diagonal)/2. On p^k all leaves are identical and the count reduces to unordered binary trees with k indistinguishable leaves, the defining recursion of the Wedderburn–Etherington numbers. □

Remark 3.7 (The OEIS entry, corrected)*.* The sequence $1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, \ldots$ of values $g_{\mathcal{B}}(n)$, $n \geq 2$, is OEIS entry A375120 (“complete binary unordered tree-factorizations”; the entry sets $a(1) = 1$ by an empty-factorization convention). Its original formula and program counted the two children as an ordered pair whenever the factor split is (d, d), a convention that disagrees with Proposition 3.6 first at $n = 144$ (giving 42 where the unordered count is 41) and disagrees with Wedderburn–Etherington on prime powers. The verification in Appendix A exposed the discrepancy, and the entry was corrected following this work in July 2026, with a table of values to $n = 10000$.

For $n = 12 = 2^2 \cdot 3$ (Figure 1) the four levels have $g_{\mathcal{T}} = C_2 \cdot \frac{3!}{2!} = 6$, $g_{\mathcal{W}} = 3$, $g_{\mathcal{B}} = 2$, $g_{\mathcal{M}} = 1$, so the entropies are $S_{\mathcal{T}}(12) = \log 6$, $S_{\mathcal{W}}(12) = \log 3$, $S_{\mathcal{B}}(12) = \log 2$, and 0 in Fock space. Table 2 lists the three nontrivial counts for small n. All formulas and recursions of this subsection were verified against brute-force enumeration for every $n \leq 20000$ (Appendix A).

n	Ω	$g_{\mathcal{T}}$	$g_{\mathcal{B}}$	$g_{\mathcal{W}}$	n	Ω	$g_{\mathcal{T}}$	$g_{\mathcal{B}}$	$g_{\mathcal{W}}$
2	1	1	1	1	20	3	6	2	3
4	2	1	1	1	24	4	20	4	4
6	2	2	1	2	27	3	2	1	1
8	3	2	1	1	28	3	6	2	3
10	2	2	1	2	30	3	12	3	6
12	3	6	2	3	32	5	14	3	1
16	4	5	2	1	36	4	30	6	6
18	3	6	2	3	64	6	42	6	1

Table 2: Degeneracies over n at the three structured levels (primes and other rows with all counts 1 omitted). Note $g_{\mathcal{T}} = C_{\Omega-1} \cdot g_{\mathcal{W}}$ throughout, and $g_{\mathcal{B}}(2^k) = 1, 1, 1, 2, 3, 6$ for $k = 1, \ldots, 6$ (Wedderburn–Etherington).

4 Partition functions: self-consistency equations and the temperature ladder

4.1 Four gases, four functional equations

Write $E(X) = \log N(X)$ and let $\beta = s$ be the inverse temperature. Let $P(s) = \sum_p p^{-s} = \sum_p e^{-s\varepsilon_p}$ denote the prime zeta function, physically the single-particle partition function of the mode spectrum $\{\log p\}$. For each corner $\mathcal{X} \in \{\mathcal{T}, \mathcal{W}, \mathcal{B}, \mathcal{M}\}$ define the canonical partition function

$$Z_{\mathcal{X}}(s) \;=\; \sum_{X \in \mathcal{X}} e^{-sE(X)} \;=\; \sum_{X \in \mathcal{X}} \frac{1}{N(X)^s} \;=\; \sum_{n \geq 2} \frac{g_{\mathcal{X}}(n)}{n^s}$$

(for $\mathcal{M}$ we include the empty configuration, of shadow 1, so that $Z_{\mathcal{M}} = \zeta$ exactly).

Theorem 4.1 (Functional equations)**.** *As identities of Dirichlet series,*

$$Z_{\mathcal{T}}(s) = P(s) + Z_{\mathcal{T}}(s)^2, \qquad Z_{\mathcal{T}} = \frac{1 - \sqrt{1 - 4P(s)}}{2} = \sum_{k \geq 1} C_{k-1}\, P(s)^k, \tag{4.1}$$

$$Z_{\mathcal{W}}(s) = P(s) + P(s)\, Z_{\mathcal{W}}(s), \qquad Z_{\mathcal{W}} = \frac{P(s)}{1 - P(s)} = \sum_{k \geq 1} P(s)^k, \tag{4.2}$$

$$Z_{\mathcal{B}}(s) = P(s) + \tfrac{1}{2}\big(Z_{\mathcal{B}}(s)^2 + Z_{\mathcal{B}}(2s)\big), \qquad Z_{\mathcal{B}} = 1 - \sqrt{1 - 2P(s) - Z_{\mathcal{B}}(2s)}, \tag{4.3}$$

$$Z_{\mathcal{M}}(s) = \zeta(s) = \prod_p \frac{1}{1 - p^{-s}}, \qquad \zeta = \exp\Big(\sum_{k \geq 1} \frac{P(ks)}{k}\Big). \tag{4.4}$$

In (4.1) *and* (4.3) *the branch of the square root is fixed by* $Z \to 0$ *as* $\operatorname{Re} s \to \infty$.

Proof. (4.1): a tree is a leaf or an ordered pair of trees, and N^{-s} is multiplicative over $\boxtimes$, so the sum over pairs factors as $Z_{\mathcal{T}}^2$. Solving the quadratic and expanding gives the Catalan series. (4.2): a word is a prime or a prime followed by a word. (4.3): an element of $\mathcal{B}$ is a leaf or an unordered pair $\{B_1, B_2\}$, possibly equal; ordered pairs contribute $Z_{\mathcal{B}}^2$, the diagonal contributes $\sum_B N(B)^{-2s} = Z_{\mathcal{B}}(2s)$, and unordered pairs with repetition are half their sum, the same Pólya computation as in Proposition 3.6. (4.4): independence of the occupation numbers gives the Euler product, and $\log \zeta(s) = \sum_p \sum_{k \geq 1} p^{-ks}/k = \sum_k P(ks)/k$. $\square$

Physical reading. Each equation is a named piece of physics. *(4.4) is the free Bose gas*: mode p with occupancy k costs energy $k \log p$, so

$$Z_{\mathcal{M}}(\beta) \;=\; \prod_p \sum_{k\geq 0} e^{-\beta k \log p} \;=\; \prod_p \frac{1}{1-p^{-\beta}} \;=\; \zeta(\beta):$$

the Euler product is the textbook free-boson partition function: the gas of numbers of Julia [23]. *(4.2) is the distinguishable-particle gas*: ordered k-particle states with no $1/k!$, the counting that produces the Gibbs paradox. *(4.1) is a self-consistency equation*: a composite is either an input particle or a combination of two composites, with energy exactly conserved, so $Z = P + Z^2$. This is the statistical bootstrap of Hagedorn–Frautschi [21, 19], here with input spectrum the primes and the combination ordered and binary. *(4.3) is the unordered version*: unordered combination, with the $Z(2s)$ term the standard exchange correction for the two branches being identical, the same mechanism that separates Bose from Boltzmann counting.

Remark 4.2 (One gas per statistics). The four right-hand sides are the four classical constructions of the symbolic method [18] (sequence, multiset, planar binary tree, unordered binary tree), each applied to the single-particle element P. Choosing a corner of the square is choosing a *statistics*: what to remember of order and of history. There is one partition function per statistics, all exact functionals of $P(s), P(2s), \ldots$, and the Riemann zeta function is the Bose member of the family. Only the multiset construction has an empty structure, which is why $n = 1$ belongs to ζ and to no other corner (Remark 2.3).

4.2 The temperature ladder

Theorem 4.3 (Inverse limiting temperatures)**.** *Each series has a finite abscissa of convergence $\sigma_{\mathcal{X}}$, determined by P:*

$$\begin{array}{llll}
\mathcal{M}: & \sigma_{\mathcal{M}} = 1 & \textit{(pole of } \zeta\textit{)}, & Z \to \infty; \\
\mathcal{W}: & P(\sigma_{\mathcal{W}}) = 1, & \sigma_{\mathcal{W}} = 1.3994333287263303\ldots, & Z \to \infty; \\
\mathcal{B}: & 2P(\sigma_{\mathcal{B}}) + Z_{\mathcal{B}}(2\sigma_{\mathcal{B}}) = 1, & \sigma_{\mathcal{B}} = 1.9884768518009081\ldots, & Z \to 1; \\
\mathcal{T}: & P(\sigma^*) = \frac{1}{4}, & \sigma^* = 2.5973851271346716\ldots, & Z \to \frac{1}{2},
\end{array}$$

and $1 < \sigma_{\mathcal{W}} < \sigma_{\mathcal{B}} < \sigma^*$: every remembered layer of structure strictly raises the critical inverse temperature.

Proof sketch. All coefficients are nonnegative, so each abscissa is located by the real singularity. For $\mathcal{W}$, the geometric series diverges exactly when $P \geq 1$. For $\mathcal{T}$, the Catalan series $\sum C_{k-1} P^k$ has radius of convergence $\frac{1}{4}$ in P; since $C_{k-1} 4^{-k} \sim k^{-3/2}/(4\sqrt{\pi})$ is summable, the series still converges at $P = \frac{1}{4}$, with value $\sum_k C_{k-1} 4^{-k} = \frac{1}{2}$. For $\mathcal{B}$, equation (4.3) defines $Z_{\mathcal{B}}$ by descending the argument-doubling cascade ($Z_{\mathcal{B}}(2s)$ lies deep in the convergent region), and the singularity occurs where the discriminant $1 - 2P(s) - Z_{\mathcal{B}}(2s)$ vanishes, at which point $Z_{\mathcal{B}} = 1$: this is Otter's criticality condition for unordered trees [32, 18]. The numerical values are computed to the stated precision in Appendix A; the strict ordering follows from $g_{\mathcal{M}} \leq g_{\mathcal{W}}, g_{\mathcal{B}} \leq g_{\mathcal{T}}$ together with strict inequality on a positive-density set of n. ☐

Physical reading. The mechanism is Hagedorn's [21]. The number of microstates at energy $E = \log n$ grows like $e^{\sigma_{\mathcal{X}} E}$ (up to powers), so the Boltzmann sum $\sum e^{(\sigma_{\mathcal{X}} - \beta)E}$ diverges for $\beta < \sigma_{\mathcal{X}}$: the gas has a maximal temperature $T_H = 1/\sigma_{\mathcal{X}}$, the *limiting temperature* (called the *Hagedorn* *temperature* in the hadron setting). Remembering order and history multiplies the microstate count at each energy, raises the entropy per unit energy, and lowers the limiting temperature:

gas	states	β_H	$T_H = 1/\beta_H$	transition
Bose	occupation numbers	1	1	pole
Boltzmann	ordered factorizations	$1.399433\ldots$	$0.714575\ldots$	pole
non-planar tree	unordered histories	$1.988477\ldots$	$0.502897\ldots$	cusp, $Z_{\mathcal{B}} \to 1$
planar tree	ordered histories	$2.597385\ldots$	$0.385003\ldots$	cusp, $Z_{\mathcal{T}} \to \frac{1}{2}$

Remark 4.4 (Pole versus cusp: is the limiting temperature attainable?). The symmetrized (associative) gases diverge at their critical point: ζ and $Z_{\mathcal{W}}$ have simple poles ($Z_{\mathcal{W}}$ because $P'(\sigma_{\mathcal{W}}) = -1.7311562\ldots \neq 0$). The history-keeping (non-associative) gases converge at theirs: expanding (4.1) at σ^*,

$$Z_{\mathcal{T}}(s) \;=\; \tfrac{1}{2} - \sqrt{|P'(\sigma^*)|}\,(s-\sigma^*)^{1/2} + O(s-\sigma^*), \qquad \sqrt{|P'(\sigma^*)|} = 0.4796402876\ldots,$$

a square-root branch point with finite value, and similarly $Z_{\mathcal{B}} \to 1$ at $\sigma_{\mathcal{B}}$. The canonical ensemble exists *at* T_H, but the mean energy

$$U(\beta) = -\frac{\partial}{\partial\beta}\log Z_{\mathcal{T}}(\beta) \;\sim\; \frac{\sqrt{|P'(\sigma^*)|}}{2\,Z_{\mathcal{T}}(\sigma^*)}\,(\beta-\sigma^*)^{-1/2} \;\to\; \infty$$

diverges as $\beta \to \sigma^{*+}$: a continuous transition with divergent susceptibility, rather than a divergent partition function. Whether the limiting temperature is attainable, the classic question of the bootstrap literature [31], is decided here by associativity alone. The square-root branch point itself is the universal singularity of tree and branched-polymer generating functions (susceptibility exponent $\frac{1}{2}$) [2], the same criticality that governs planar-diagram counting in the Gaussian matrix model, where the Catalan numbers arise as the planar moments and $\frac{1}{4}$ is the edge of the Wigner semicircle [4].

Remark 4.5 (The classical bootstrap on the prime spectrum). The historical statistical bootstrap model combines in one shot: a composite is an input particle *or an unordered cluster of* $k \geq 2$ *composites*, Boltzmann-weighted by $1/k!$. Its generating equation is $G = \varphi + (e^G - 1 - G)$, whose square-root singularity, first remarked by Nahm [31], occurs when the input function reaches the universal value

$$\varphi_c \;=\; 2\log 2 - 1 \;=\; 0.3862943611\ldots, \qquad G_c = \log 2.$$

Evaluated on the empirical hadron spectrum, this criticality condition is what fixed the original $T_H \approx 160\,\mathrm{MeV}$. Transplanted onto the prime spectrum ($\varphi = P$), it gives the inverse Hagedorn temperatures

$$\beta_H^{\mathrm{Boltz}} = 2.1487511659\ldots, \qquad \beta_H^{\mathrm{Bose}} = 2.3072314551\ldots,$$

the Bose value arising from full multiset statistics ($e^G \mapsto \exp\sum_{k\geq 1} G(ks)/k$, with criticality when $P(s) + 2\sum_{k\geq 2} G(ks)/k$ reaches $2\log 2 - 1$; see Appendix A). These classical gases slot into the same family as the four corners (same input spectrum, same square-root mechanism) but sit off the ladder of Theorem 4.3 because their symmetry-factor weights are fractional rather than integer counts. In bootstrap language, the tree world $\mathcal{T}$ is a bootstrap model whose decay chains are fully resolved: an integer's prime factorization is its ultimate decay inventory, and the primes are the stable particles.

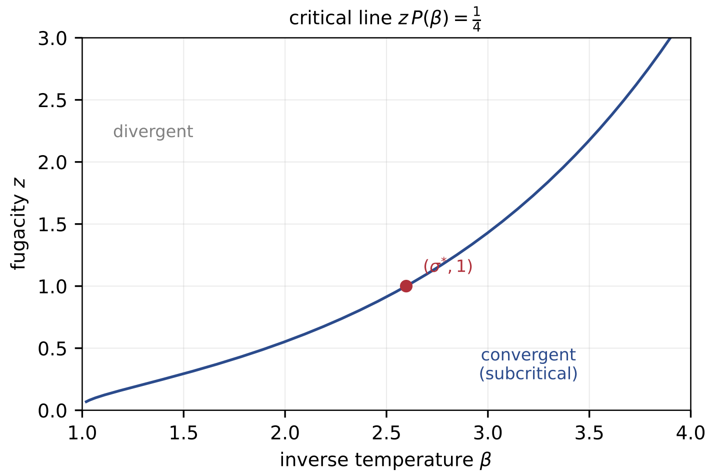

Figure 2: Phase diagram of the planar tree gas with a fugacity z per particle (Remark 4.6). The grand partition function converges below the critical line $z\,P(\beta) = \frac{1}{4}$ and diverges above it. The tree world $\mathcal{T}$ is the slice $z = 1$, meeting the line at the limiting temperature σ^*; raising z pushes the limiting temperature up without bound.

Remark 4.6 (Fugacity and the phase diagram). Give each particle (each leaf) a fugacity z, that is, weight a tree by $z^{|T|}$. The grand partition function of the planar tree gas, $Z_{\mathcal{T}}(s,z) = \sum_T z^{|T|} N(T)^{-s}$, satisfies the same self-consistency equation with a rescaled input, since a tree is a leaf (weight $z\,p^{-s}$) or a pair:

$$Z_{\mathcal{T}}(s,z) = z\,P(s) + Z_{\mathcal{T}}(s,z)^2, \qquad Z_{\mathcal{T}}(s,z) = \frac{1-\sqrt{1-4z\,P(s)}}{2}.$$

The limiting temperature becomes z-dependent: convergence holds up to the *critical line*

$$z\,P(\beta) = \tfrac{1}{4}$$

in the (β, z) quarter-plane, which passes through $(\sigma^*, 1)$ and, as z grows, sweeps the limiting temperature to arbitrarily high values (Figure 2). The tree world $\mathcal{T}$ is the slice $z = 1$. Fugacity thus turns the single Hagedorn point of §4 into a one-parameter family, and the same substitution $P \mapsto zP$ carries every result of §5 along the line: the complex-temperature zeros of the grand gas are the solutions of $z\,P(s) = \frac{1}{4}$, a level set of P at a tunable height. The physical zeros ($z = 1$, level $\frac{1}{4}$) are one horizontal cut through the value-distribution theory of P developed there.

Remark 4.7 (Growth of the counting functions). By Wiener–Ikehara applied to $Z_{\mathcal{W}}$,

$$\sum_{n\le x} g_{\mathcal{W}}(n) \sim \frac{x^{\sigma_{\mathcal{W}}}}{\sigma_{\mathcal{W}}|P'(\sigma_{\mathcal{W}})|} \approx 0.413\,x^{1.399\ldots},$$

while for the tree level transfer heuristics for the square-root singularity suggest $\sum_{n\le x} g_{\mathcal{T}}(n) \asymp x^{\sigma^*}(\log x)^{-3/2}$ (Problem 8.2). The counts can be extreme: along $n = 2^k$, $g_{\mathcal{T}}(n) = C_{k-1} \asymp n^2/(\log n)^{3/2}$, so the number of microstates over a single integer can approach its square. (Kalmár's classical count [25] of ordered factorizations into arbitrary parts ≥ 2, exponent $\rho = 1.7286\ldots$ with $\zeta(\rho) = 2$, sits between the Boltzmann and non-planar rungs but weights composite parts.)

5 Complex-temperature zeros of the arithmetic gas

Lee and Yang taught that the thermodynamics of a system is encoded in the complex singularities of its partition function, in complex fugacity for them and in complex temperature for Fisher [27, 16]; the latter are the *Fisher zeros.* We now locate the complex-temperature singularities of the tree gas. Since $Z_{\mathcal{T}} = \frac{1}{2}(1 - \sqrt{1-4P})$ is an *algebraic* function of P, they come in exactly two kinds: those inherited from the single-particle input P, and the new branch points where the discriminant vanishes.

5.1 What the gas inherits: the Riemann zeros

The prime zeta function continues to the right half-plane by Möbius inversion of (4.4):

$$P(s) \;=\; \sum_{k\geq 1} \frac{\mu(k)}{k} \log\zeta(ks), \qquad \operatorname{Re} s > 0, \tag{5.1}$$

valid off the singular set. Equation (5.1) shows that P has logarithmic branch points at $s = 1/k$ (from the pole of ζ) *and at $s = \rho/k$ for every nontrivial zero ρ of ζ*; these points cluster on the imaginary axis, which is a natural boundary (Landau–Walfisz [26]). Two consequences:

1. Lifting arithmetic to trees does not dissolve the Riemann Hypothesis. The zeros of ζ reappear, relocated, inside $Z_{\mathcal{T}}$ as branch points at ρ/k. Where the Hilbert–Pólya program seeks a Hamiltonian whose *spectrum* is the Riemann zeros [3], the arithmetic gas offers the complementary picture: the zeros enter as singularities of the single-particle input, while the gas contributes its own critical singularities on top, and those are computable.

2. No one-variable partition function of this family can contain more analytic information than P itself. Genuinely new invariants must either weight the *shape* of histories (two-variable partition functions, Problem 8.4) or couple to the additive world (§6).

5.2 The new singularities: tree zeros

The new singularities of $Z_{\mathcal{T}}$ are the solutions of

$$P(s) = \tfrac{1}{4}, \tag{5.2}$$

square-root branch points of $Z_{\mathcal{T}}$: the complex-temperature zeros of the tree gas. We call them *tree zeros* (they are the zeros of the discriminant $1-4P$). On the real axis there is exactly one, σ^*, since P is strictly decreasing there. Off the axis:

Lemma 5.1. *For $\sigma > 0$ where $P(\sigma)$ converges and $t \neq 0$: $|P(\sigma+it)| < P(\sigma)$.*

Proof. $|\sum_p p^{-\sigma} p^{-it}| \leq \sum_p p^{-\sigma}$ with equality iff all phases $t \log p$ agree modulo 2π. For $t \neq 0$ this fails already for $p = 2, 3$ since $\log 3/\log 2$ is irrational. □

Proposition 5.2. *Every non-real tree zero satisfies* $\operatorname{Re} s < \sigma^*$.

Proof. If $\sigma \geq \sigma^*$ and $t \neq 0$ then $|P(\sigma+it)| < P(\sigma) \leq P(\sigma^*) = \frac{1}{4}$. □

k	$\operatorname{Re} s$	$\operatorname{Im} s$	$t/\frac{2\pi}{\log 2}$
1	1.54427014392	10.0122461654	1.105
2	2.14220467429	17.8224730193	1.966
3	2.24736156780	27.6431435747	3.050
4	2.15502047591	35.5407136013	3.921
5	2.39048394209	45.5394484588	5.024

Table 3: All solutions of $P(s) = \frac{1}{4}$ with $1.4 < \operatorname{Re} s < 2.7$, $0 < \operatorname{Im} s < 50$ (each accompanied by its complex conjugate). A coarser sweep of $1.05 < \operatorname{Re} s < 1.4$ found no further solutions in this height range.

The tree zeros with $0 < t < 50$ are listed in Table 3 (method in Appendix A).

The ordinates stay near integer multiples of $2\pi/\log 2 = 9.0647\ldots$: the partition function is almost periodic in $\operatorname{Im}\beta$, with the recurrence frequency set by the lightest mode, $\varepsilon_2 = \log 2$, whose Boltzmann factor $2^{-\beta}$ dominates P; interference from $3^{-\beta}$ and the later modes shifts each singularity, strongly so for $k = 1$. The real parts, on the other hand, wander. A tree analogue of the Riemann Hypothesis, with all zeros on one vertical line, is therefore false, and the right question is distributional: with what density do the zeros fill the strip? Theorem 5.3 answers this.

5.3 The density of tree zeros

Within the half-plane of almost periodicity the tree zeros have an exact asymptotic density, computable from a random-phase model of the gas.

Theorem 5.3 (Density of the complex-temperature zeros)**.** *For $\sigma > \frac{1}{2}$ let $X_\sigma = \sum_p p^{-\sigma} e^{i\theta_p}$ with (θ_p) independent and uniform on $[0, 2\pi)$ (the series converges almost surely and in L^2, since $\sum_p p^{-2\sigma} = P(2\sigma) < \infty$), and define the* Jensen function *and the* dominance probabilities

$$\Phi(\sigma) = \mathbb{E}\,\log\big|X_\sigma - \tfrac{1}{4}\big|, \qquad q_p(\sigma) = \mathbb{P}\Big(\,\big|X_\sigma - p^{-\sigma} e^{i\theta_p} - \tfrac{1}{4}\big| < p^{-\sigma}\Big).$$

(i) For every $\sigma > 1$, $\frac{1}{T}\int_0^T \log|P(\sigma + it) - \frac{1}{4}|\,dt \to \Phi(\sigma)$ as $T \to \infty$; Φ is convex, and $\Phi(\sigma) = \log\frac{1}{4}$ for $\sigma \geq \sigma^*$.

(ii) For $1 < \sigma_1 < \sigma_2$, the number $N_{[\sigma_1,\sigma_2]}(T)$ of tree zeros with real part in $[\sigma_1, \sigma_2]$ and $0 < t \leq T$ satisfies

$$\frac{N_{[\sigma_1,\sigma_2]}(T)}{T} \longrightarrow \frac{\Phi'(\sigma_2) - \Phi'(\sigma_1)}{2\pi}.$$

(iii) $\Phi'(\sigma) = -\sum_p (\log p)\, q_p(\sigma)$. In particular the density of tree zeros to the right of the line $\operatorname{Re} s = \sigma_1$ is $\frac{1}{2\pi}\sum_p (\log p)\, q_p(\sigma_1)$, and $q_p(\sigma) = 0$ for $\sigma \geq \sigma^*$, reproving Proposition 5.2.

The proof rests on two regularity lemmas for the random model and one imported estimate from the theory of almost periodic functions. Throughout, Y_σ denotes either X_σ or a *deleted* sum $X_\sigma - p^{-\sigma} e^{i\theta_p}$, and f_σ denotes its density.

Lemma 5.4 (Regularity of the random model)**.** *Fix $\frac{1}{2} < \sigma_0 \le \sigma_1 < \infty$. Uniformly for $\sigma \in [\sigma_0, \sigma_1]$: (a) Y_σ has a C^∞ density f_σ on $\mathbb{C}$ with $\sup f_\sigma \le C(\sigma_0, \sigma_1)$; (b) $\mathbb{E}\,|Y_\sigma - w|^{-\gamma} \le C'(\sigma_0, \sigma_1, \gamma)$ for every $w \in \mathbb{C}$ and every $0 < \gamma < 2$; (c) $\sigma \mapsto f_\sigma$ is continuous in the uniform norm, and consequently each q_p is continuous on $(\frac{1}{2}, \infty)$.*

Proof. (a) A single mode $ae^{i\theta}$ with θ uniform has characteristic function $\mathbb{E}\, e^{i\langle \xi,\, ae^{i\theta}\rangle} = J_0(a|\xi|)$, so by independence $\widehat{f_\sigma}(\xi) = \prod_q J_0(q^{-\sigma}|\xi|)$ over the participating primes. With the classical bound $|J_0(x)| \le \min(1, C_0 x^{-1/2})$ and any K fixed primes $q \le Q$ (all primes participate except possibly the one deleted mode),

$$|\widehat{f_\sigma}(\xi)| \;\le\; C_0^K\, Q^{K\sigma_1/2}\, |\xi|^{-K/2} \qquad (|\xi| \ge 1).$$

Taking $K = 5$ makes $\widehat{f_\sigma}$ integrable on $\mathbb{R}^2$ with a bound uniform in σ, so Fourier inversion yields a continuous density with $\sup f_\sigma \le (2\pi)^{-2}\|\widehat{f_\sigma}\|_{L^1}$; taking K arbitrarily large gives integrability of $|\xi|^m \widehat{f_\sigma}$ for every m, hence smoothness. (b) Split at distance 1: $\mathbb{E}|Y_\sigma - w|^{-\gamma} \le \sup f_\sigma \int_{|z|\le 1} |z|^{-\gamma}\, dz + 1 < \infty$ since $\gamma < 2$. (c) As $\sigma' \to \sigma$ the characteristic functions converge pointwise (each factor does, and the infinite product converges locally uniformly by the L^2 tail bound), dominated by the integrable envelope of (a); Fourier inversion turns this into uniform convergence of the densities. Then $q_p(\sigma) = \int_{|w-1/4|<p^{-\sigma}} f_\sigma(w)\, dw$, taken with the deleted density, is continuous, the radius being continuous as well. □

Lemma 5.5 (Dominance bounds)**.** *Uniformly for $\sigma \ge 1$, $q_p(\sigma) \le \pi C\, p^{-2\sigma}$. Hence $\sum_p (\log p)\, q_p(\sigma)$ converges uniformly on $[1, \infty)$ and is continuous there.*

Proof. The dominance event puts the deleted sum in a disk of radius $p^{-\sigma}$, so $q_p \le \sup f_\sigma \cdot \pi p^{-2\sigma}$, with Lemma 5.4(a) applied on $[1, \sigma^* + 1]$; for $\sigma > \sigma^*$ one has $q_p = 0$ outright, as shown in step (i) below. Since $\sum_p (\log p)\, p^{-2} < \infty$, Weierstrass gives uniform convergence, and continuity follows from Lemma 5.4(c). □

Lemma 5.6 (Small values; Jessen–Tornehave)**.** *Let f be uniformly almost periodic and analytic on a closed substrip of* $\operatorname{Re} s > 1$*, not identically zero. Then for every interior σ,*

$$\lim_{\varepsilon \downarrow 0}\, \limsup_{T\to\infty}\, \frac{1}{T}\int_0^T \log^+ \frac{\varepsilon}{|f(\sigma + it)|}\, dt \;=\; 0.$$

This is Theorem 5 of Jessen–Tornehave [22]: the means of $\log\max(|f(\sigma+it)|, \varepsilon)$ *converge, uniformly on substrips, to the Jensen function as $\varepsilon \to 0$, which is exactly the displayed statement. The same estimate, for ζ and its a-points, is carried out by Borchsenius and Jessen [11].*

Proof of Theorem 5.3. On a closed substrip of $\operatorname{Re} s > 1$ the series $P(s) - \frac{1}{4}$ converges absolutely and uniformly, so it is a uniformly almost periodic analytic function with frequency set $\{\log p\}$. The frequencies are linearly independent over $\mathbb{Q}$: a rational relation $\sum_k c_k \log p_k = 0$ would give $\prod_k p_k^{c_k} = 1$, impossible by unique factorization.

(i). For $y \ge 2$ and $\varepsilon > 0$ put $P_y(s) = \sum_{p \le y} p^{-s}$, let $X_{\sigma,y}$ be the correspondingly truncated model, and let $\ell_\varepsilon(z) = \log\max(|z|, \varepsilon)$, which is $(1/\varepsilon)$-Lipschitz in $|z|$. The function $t \mapsto \ell_\varepsilon(P_y(\sigma + it) - \frac{1}{4})$ is a continuous function of the finitely many phases $(t \log p \bmod 2\pi)_{p \le y}$, which equidistribute on their torus by Weyl's criterion [40] and the linear independence above; hence

its time average converges to $\mathbb{E}\,\ell_\varepsilon(X_{\sigma,y}-\frac14)$. Since $|P-P_y|\le\sum_{p>y}p^{-\sigma}\to 0$ uniformly in t and $\mathbb{E}|X_\sigma-X_{\sigma,y}|\to 0$, letting $y\to\infty$ gives, for every $\varepsilon>0$,

$$\lim_{T\to\infty}\frac{1}{T}\int_0^T \ell_\varepsilon\big(P(\sigma+it)-\tfrac14\big)\,dt \;=\; \mathbb{E}\,\ell_\varepsilon\big(X_\sigma-\tfrac14\big).$$

As $\varepsilon\downarrow 0$ the right side decreases to $\Phi(\sigma)$, finite by Lemma 5.4(b) with $\gamma=1$; on the left, $\ell_\varepsilon-\log|\cdot|=\log^+(\varepsilon/|\cdot|)$ is controlled by Lemma 5.6. Together these give $\frac1T\int_0^T\log|P(\sigma+it)-\frac14|\,dt\to\Phi(\sigma)$, the value-distribution mechanism of Bohr and Jessen for $\log\zeta$ [10]. Convexity of Φ is inherited from the finite-T means, which are convex in σ because $\log|P-\frac14|$ is subharmonic.

For $\sigma>\sigma^*$, fix any prime q and condition on the other phases: writing $X_\sigma-\frac14=W_q+q^{-\sigma}e^{i\theta_q}$, the pointwise bound $|W_q|\ge\frac14-(P(\sigma)-q^{-\sigma})>q^{-\sigma}$ holds because $P(\sigma)<\frac14$, and the circular mean $\frac{1}{2\pi}\int_0^{2\pi}\log|W+ae^{i\theta}|\,d\theta=\log\max(|W|,a)$ gives $\mathbb{E}\log|X_\sigma-\frac14|=\mathbb{E}\log|W_q|$: deleting one mode does not change the Jensen function. Iterating over all $p\le y$ and letting $y\to\infty$, the remaining tail T_y satisfies $|T_y-\frac14|\in[\frac14-\delta_y,\frac14+\delta_y]$ with $\delta_y\to 0$, so $\Phi(\sigma)=\lim_y\mathbb{E}\log|T_y-\frac14|=\log\frac14$. The same pointwise bound shows the dominance event is empty, $q_p(\sigma)=0$; at $\sigma=\sigma^*$ both statements follow by continuity (Lemmas 5.4(c) and 5.5, and convexity for Φ).

(ii) is the Jensen formula for almost periodic functions of Jessen–Tornehave (Theorem 7 and §46 of [22]), applied to $P-\frac14$: when the Jensen function is differentiable at σ_1 and σ_2, the relative frequency of zeros in the strip (σ_1,σ_2) exists and equals $\big(\varphi'(\sigma_2)-\varphi'(\sigma_1)\big)/2\pi$. By (i) the Jensen function is Φ, and by (iii) below Φ' exists and is continuous on $(1,\infty)$, so there are no exceptional points.

(iii). Pathwise, $\sigma\mapsto X_\sigma$ is smooth for $\sigma>1$ with $|\partial_\sigma X_\sigma|\le\sum_p(\log p)\,p^{-\sigma}=|P'(\sigma)|$, an absolutely convergent series. For $h\neq 0$ the difference quotients obey

$$\Big|\frac{\log|X_{\sigma+h}-\frac14|-\log|X_\sigma-\frac14|}{h}\Big| \;\le\; \sup_{|u|\le|h|}|P'(\sigma+u)|\cdot\Big(\big|X_\sigma-\tfrac14\big|^{-1}\wedge\big|X_{\sigma+h}-\tfrac14\big|^{-1}\Big)^*$$

in the elementary form $|\log x-\log y|\le|x-y|/\min(x,y)$; by Lemma 5.4(b) with $\gamma=\frac32$ the family of quotients is bounded in $L^{3/2}$, hence uniformly integrable, and it converges almost surely (the event $X_\sigma=\frac14$ is null). Vitali's theorem then allows differentiation under the expectation:

$$\Phi'(\sigma)\;=\;\mathbb{E}\;\mathrm{Re}\,\frac{\partial_\sigma X_\sigma}{X_\sigma-\frac14}\;=\;-\sum_p(\log p)\;\mathbb{E}\;\mathrm{Re}\,\frac{p^{-\sigma}e^{i\theta_p}}{X_\sigma-\frac14},$$

the exchange of sum and expectation being dominated by $|P'(\sigma)|\,|X_\sigma-\frac14|^{-1}$. Average over θ_p first, holding the other phases fixed: for W with $|W|\neq a$,

$$\frac{1}{2\pi}\int_0^{2\pi}\frac{ae^{i\theta}}{W+ae^{i\theta}}\,d\theta\;=\;\begin{cases}1 & |W|<a,\\ 0 & |W|>a,\end{cases}$$

by residues, and the boundary event $|W|=a$ has probability zero because the deleted sum has a density (Lemma 5.4(a)). This yields $\Phi'(\sigma)=-\sum_p(\log p)\,q_p(\sigma)$, real as it must be, and continuous by Lemma 5.5. ☐

Corollary 5.7 (Total density)**.** *Set* $D(\sigma)=\frac{1}{2\pi}\sum_p(\log p)\,q_p(\sigma)$*. Then* D *is finite and continuous on* $[1,\infty)$ *and vanishes for* $\sigma\ge\sigma^*$*; for every* $\sigma_1>1$ *the tree zeros with* $\mathrm{Re}\,s>\sigma_1$ *have asymptotic*

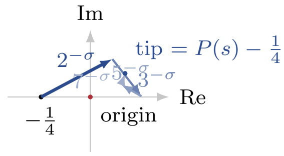

Figure 3: The value $P(s) - \frac{1}{4}$ at $s = \sigma + it$ as a chain of vectors from $-\frac{1}{4}$, the p-th of length $p^{-\sigma}$ rotating at speed $\log p$ as t runs. A tree zero is a moment the tip hits the origin. When one hand ($2^{-\sigma}$ here) exceeds the anchored sum of the rest, it alone controls the winding and forces one zero per turn; the density of zeros is the total rate of such dominance, $\frac{1}{2\pi}\sum_p (\log p)\, q_p$ (Theorem 5.3).

density $D(\sigma_1)$; and as $\sigma_1 \downarrow 1$ these densities increase to the finite limit

$$D(1) \;=\; \frac{1}{2\pi}\sum_p (\log p)\, q_p(1) \;=\; 0.1436\ldots,$$

which we call the total density *of the tree zeros in* $\operatorname{Re} s > 1$ *(Appendix A).*

Proof. Finiteness and continuity on $[1,\infty)$ are Lemma 5.5; note that the random model, and with it q_p and D, is defined for all $\sigma > \frac{1}{2}$, so $\sigma = 1$ is an interior point for the model even though almost periodicity fails there. For $\sigma_1 > 1$ the zeros right of σ_1 all lie in the substrip $(\sigma_1, \sigma^*]$ by Proposition 5.2, so Theorem 5.3(ii)–(iii) gives their density as $D(\sigma_1) - D(\sigma^* + 1) = D(\sigma_1)$. These densities are non-decreasing as σ_1 decreases, by nesting, and converge to $D(1)$ by continuity. Whether the zero count in the closed half-plane $\operatorname{Re} s > 1$ is exactly linear with slope $D(1)$ depends on how the zeros accumulate toward the boundary line, where the continuation (5.1) takes over; that is part of Problem 8.1. $\square$

Physical reading. Fix σ and let t run: $P(\sigma + it) - \frac{1}{4}$ is a chain of clock hands anchored at $-\frac{1}{4}$, one per prime, the p-hand of length $p^{-\sigma}$ turning at angular speed $\log p$ (Figure 3). A tree zero is a moment when the tip reaches the origin. Whenever one hand is longer than the anchored sum of all the others it drags the tip once around the origin per revolution, forcing one zero: this is the event q_p measures. At $\sigma = 1.30$ the hands have lengths $2^{-1.3} = 0.41$, $3^{-1.3} = 0.24$, $5^{-1.3} = 0.12, \ldots$, and $q_2 = 0.624$, $q_3 = 0.148$, $q_5 = 0.042$, $q_7 = 0.017$; weighted by speed the lightest hand accounts for only 59% of the zeros, which is why the density exceeds the naive one-per-period guess $\log 2/2\pi$. This is exactly the classical mean-motion problem of celestial mechanics: a hand exceeding the anchored sum of the rest is what Lagrange called a *preponderant term* (see the introduction of [22]), and the Jessen–Tornehave theory is the problem's solution.

The Monte Carlo dominance probabilities give a curve $D(\sigma)$ rising from $D(\sigma^*) = 0$ to

$$D(1.30) = 0.1162, \quad D(1.10) = 0.1314, \quad D(1.05) = 0.1372, \quad D(1) = 0.1436$$

(Appendix A), the last value being the total density of Corollary 5.7. Against this, a full census to height 5000 (Figure 4) locates 595 tree zeros, all with $1.26 < \operatorname{Re} s < 2.57$, hence all left of σ^* as Proposition 5.2 requires. The counting function $N(T)$ tracks the predicted line $D(1)\,T$ closely (Figure 4, right); the small deficit comes from the zeros with $1 < \operatorname{Re} s < 1.26$ outside the census box, whose density $D(1) - D(1.26) \approx 0.014$ accounts for it. The mean spacing is 8.39 against the lightest-mode period $2\pi/\log 2 = 9.06$. The left edge drifts as before: the lowest real part found is 1.262 within the box, and a coarse sub-box sweep continues to turn up zeros with real parts approaching 1, where almost periodicity fails and the continuation (5.1), with its Riemann-zero branch points, takes over. What happens on and past that line is Problem 8.1.

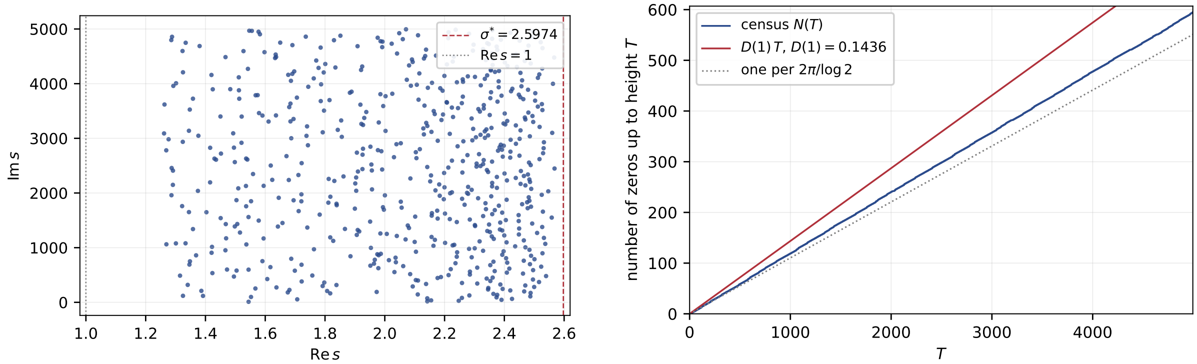

Figure 4: Left: the 595 tree zeros (solutions of $P(s) = \frac{1}{4}$) with $0 < \operatorname{Im} s < 5000$, all left of σ^* (Proposition 5.2), the real parts wandering across the strip. Right: the counting function against the predicted line $D(1)\,T$ with $D(1) = 0.1436$ (Corollary 5.7) and against the naive one-per-period line; the deficit is the tail of zeros with $\operatorname{Re} s < 1.26$ outside the census box.

5.4 Spectral statistics of the tree zeros

The 595 ordinates are numerous enough to ask how the zeros are spaced, the question one asks of any spectrum. Normalizing to unit mean spacing, the nearest-neighbour distribution is neither Poissonian (uncorrelated levels) nor of random-matrix type: it is far more rigid than both (Figure 5). The mass piles up near $s = 1$; the coefficient of variation of the raw gaps is 0.259, against 1 for a Poisson process; the smallest normalized gap observed is 0.79, and only 8% of gaps fall below $s = \frac{1}{2}$ (a Poisson process would have 39%). The number variance saturates rather than growing linearly,

$$\Sigma^2(1) = 0.18, \quad \Sigma^2(5) = 0.57, \quad \Sigma^2(10) = 0.67 \qquad (\text{Poisson: } \Sigma^2(L) = L),$$

so the count of zeros in a long window is almost deterministic. This is the expected fingerprint of the zeros of an almost periodic analytic function: a quasi-periodic skeleton (here paced by $2\pi/\log 2$) dressed with defects where a wandering lighter-mode resonance displaces a zero. It is not the GUE statistics conjectured for the Riemann zeros themselves [30]: although the tree gas inherits the Riemann zeros through P, its spectrum has entirely different rigidity. Making the empirical rigidity (near-crystalline order with a controlled density of defects) into a theorem is Problem 8.1(c).

6 The additive bridge: states, processes, and classical observables

The additive world of the theory is flat: $\mathcal{A} = \{n\,\ell : n \in \mathbb{N}_{\geq 1}\}$, heaps of a unit quantum ℓ ("one liter"), with $+\colon \mathcal{A} \times \mathcal{A} \to \mathcal{A}$ ordinary accumulation. The question left open so far is how $\mathcal{T}$ and $\mathcal{A}$ interact. The answer proposed here is that they do not interact through a map $\mathcal{T} \to \mathcal{A}$: multiplication acts on quantities. Trees are processes, and the output of the action is a third kind of object, a state of clustered matter.

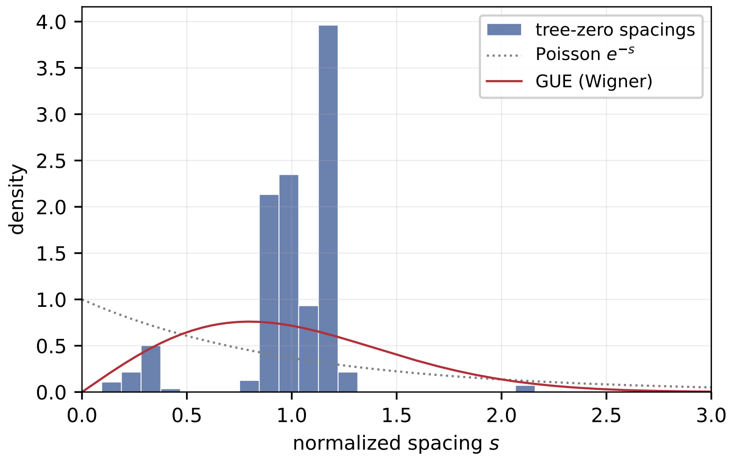

Figure 5: Nearest-neighbour spacings of the tree-zero ordinates, normalized to unit mean, against the Poisson and GUE (Wigner) laws. The distribution is more rigid than either: negligible mass below $s = \frac{1}{2}$, a sharp peak near $s = 1$, and a number variance that saturates ($\Sigma^2(10) = 0.67$ against 10 for Poisson). This is the signature of the zeros of an almost periodic function rather than of a random-matrix ensemble.

6.1 States: grouped quantities

Definition 6.1 (Grouped quantities)**.** Define *packages* t and *quantities* f by the mutual grammar

$$t \ ::= \ \ell \ \mid \ \boxed{f}, \qquad\qquad f \ ::= \ t_1 \oplus t_2 \oplus \cdots \oplus t_m \quad (m \geq 1),$$

where $\boxed{f}$ is a box containing the quantity f, and $\oplus$ is concatenation of finite sequences of packages. Let $\mathcal{G}$ denote the set of quantities. For $k \geq 2$ define the *uniform grouping*

$$k \odot f \ = \ \underbrace{\boxed{f} \oplus \boxed{f} \oplus \cdots \oplus \boxed{f}}_{k}.$$

The *flattening* $\alpha\colon \mathcal{G} \to \mathcal{A}$ counts atoms: $\alpha(\ell) = \ell$, $\alpha(\boxed{f}) = \alpha(f)$, $\alpha(f \oplus g) = \alpha(f) + \alpha(g)$. We identify $\mathcal{A}$ with $(\mathbb{N}_{\geq 1}, +)$ and write $\alpha(f) \in \mathbb{N}_{\geq 1}$.

Physical reading. A quantity is matter with a cluster hierarchy: quanta in bottles, bottles in cartons; nucleons in nuclei, nuclei in atoms. $\oplus$ places systems side by side; α is the extensive observable, total particle number, blind to packaging. $\mathcal{G}$ refines $\mathcal{A}$: a state remembers how it is packed. ($\oplus$ is associative by construction, being concatenation; one may further quotient $\mathcal{G}$ by reordering of packages, and nothing below changes, as the proofs never use the order.)

Definition 6.2 (The action)**.** Define $\rho\colon \mathcal{T} \to \mathrm{Maps}(\mathcal{G}, \mathcal{G})$ by

$$\rho(L_p)(f) = p \odot f, \qquad \rho(T_1 \boxtimes T_2) = \rho(T_1) \circ \rho(T_2).$$

Example 6.3. $\rho(L_3)(\ell) = \boxed{\ell}\boxed{\ell}\boxed{\ell}$: three bottles of one liter. Then

$$\rho(L_2 \boxtimes L_3)(\ell) = 2 \odot (\boxed{\ell}\boxed{\ell}\boxed{\ell}) = \boxed{\boxed{\ell}\boxed{\ell}\boxed{\ell}}\,\boxed{\boxed{\ell}\boxed{\ell}\boxed{\ell}},$$

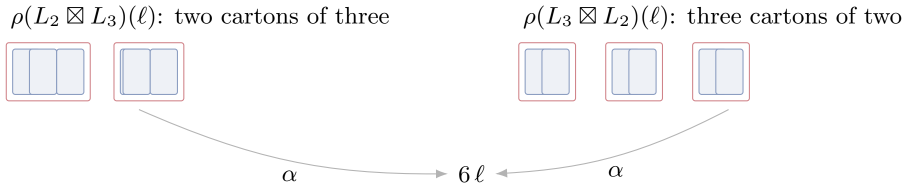

Figure 6: A tree acts on quantities as a process rather than a number. $L_2 \boxtimes L_3$ and $L_3 \boxtimes L_2$ package one liter differently: two cartons of three bottles versus three cartons of two. As states they differ (associativity would too); only the extensive count α collapses both to 6ℓ. "You never get six liters from multiplication" (§2) is Theorem 6.6: the laws hold in expectation but not on states.

two cartons each holding three bottles; whereas $\rho(L_3 \boxtimes L_2)(\ell)$ is three cartons each holding two bottles (Figure 6). Neither is "six liters"; both flatten to 6ℓ. The slogan of §1 is now a statement about the image of ρ: multiplication outputs clustered states, never flat ones.

6.2 The stratification theorems

The three classical laws now separate into three different layers of observability.

Theorem 6.4 (Associativity is additively invisible, exactly)**.** *The action ρ factors through the associativity quotient $\mathcal{T} \to \mathcal{W}$, and the induced map $\mathcal{W} \to \mathrm{Maps}(\mathcal{G}, \mathcal{G})$ is injective. Consequently the congruence $\{(T, T') : \rho(T) = \rho(T')\}$ is precisely $\approx_a$.*

Proof. Factoring: $\mathrm{Maps}(\mathcal{G}, \mathcal{G})$ is a monoid under composition, composition is associative, and ρ sends $\boxtimes$ to $\circ$; hence $\rho((T_1 \boxtimes T_2) \boxtimes T_3) = \rho(T_1) \circ \rho(T_2) \circ \rho(T_3) = \rho(T_1 \boxtimes (T_2 \boxtimes T_3))$, and $\rho(T)$ depends only on $\mathrm{word}(T)$.

Injectivity: let $w = p_1 p_2 \cdots p_k$ and evaluate on the single quantum: $F = \rho(w)(\ell) = p_1 \odot \rho(p_2 \cdots p_k)(\ell)$ consists of exactly p_1 identical top-level packages whose common content is $\rho(p_2 \cdots p_k)(\ell)$. So from F one reads off p_1 (the number of top-level packages) and recurses on the content; the recursion terminates at the boxless quantum ℓ. Distinct words therefore give distinct maps. □

Physical reading. Why does associativity feel inevitable, when nothing in §2 imposes it? Because processes compose: any semantics that interprets multiplication as something one does (anything of the form $\boxtimes \mapsto \circ$ into an endomorphism monoid) must identify at least $\approx_a$, since composition of maps is associative whether we like it or not. Our semantics identifies exactly $\approx_a$: nothing more is lost. Interaction history is unobservable in any operational reading of multiplication; to observe it, one would have to make states out of the processes themselves (currying), which is precisely to leave the additive world. By contrast, nothing forces the process monoid to be commutative, and indeed it is not.

Theorem 6.5 (Commutativity survives until expectation)**.** *For all $T \in \mathcal{T}$ and $f \in \mathcal{G}$:*

$$\alpha\big(\rho(T)(f)\big) \;=\; N(T) \cdot \alpha(f).$$

Hence the induced action on flat quantities $\mathcal{A}$ is multiplication by the shadow, and two trees act identically on $\mathcal{A}$ iff $N(T) = N(T')$, i.e. (by Proposition 3.2) iff $T \approx_{ac} T'$. In particular

$\rho(L_2 \boxtimes L_3) \neq \rho(L_3 \boxtimes L_2)$ *as maps of* $\mathcal{G}$*, yet both have the same expectation: every observer reading only particle number sees multiplication by* 6*.*

Proof. Induction: $\alpha(p \odot f) = p\,\alpha(f)$ since boxes are α-transparent, and $\alpha(\rho(T_1 \boxtimes T_2)(f)) = N(T_1)\,\alpha(\rho(T_2)(f)) = N(T_1)N(T_2)\,\alpha(f)$. The characterization of equal flat actions is unique factorization. Distinctness on $\mathcal{G}$ is Theorem 6.4 (different words). □

The two observation layers thus recover the two axes of the coarse-graining square: clustered observation quotients by exactly associativity, and extensive observation quotients by exactly associativity and commutativity; the second "exactly" is again the Fundamental Theorem of Arithmetic.

Theorem 6.6 (Distributivity holds only in expectation)**.** *For every* $T \in \mathcal{T}$ *and* $f, g \in \mathcal{G}$*:*

$$\rho(T)(f \oplus g) \;\neq\; \rho(T)(f) \oplus \rho(T)(g) \quad \text{in } \mathcal{G}, \qquad \text{yet} \qquad \alpha\big(\rho(T)(f \oplus g)\big) \;=\; \alpha\big(\rho(T)(f) \oplus \rho(T)(g)\big).$$

Proof. Let p_1 be the first letter of word(T). The left state consists of p_1 top-level packages; the right state consists of $2p_1$; since $p_1 \geq 2$ these differ. Flattening both sides gives $N(T)(\alpha f + \alpha g)$ by Theorem 6.5. □

"Process the merged system" and "process the parts, then merge" are genuinely different protocols with the same extensive outcome: the informal claim of §1, now a theorem. Note also the built-in one-sidedness: there is no $(f \oplus g) \cdot T$ at all, because states are not processes. In Loday's arithmetree [28], where both arguments are trees, distributivity likewise survives on one side only.

law	in $\mathcal{T}$ (microstates)	on states $\mathcal{G}$	in expectation (α)
associativity	fails (free)	**holds**	holds
commutativity	fails (free)	fails	**holds**
distributivity	not formulable	fails	**holds**

6.3 Where the primes come from

The theory so far imports its particle spectrum from classical arithmetic. The bridge removes this dependency: the primes can be found inside $\mathcal{G}$.

Definition 6.7. Call $n \in \mathbb{N}_{\geq 2}$ *groupable* if some quantity consisting of $k \geq 2$ identical packages, each containing $m \geq 2$ quanta, flattens to n, that is, if $n = km$ with $k, m \geq 2$. Call n *prime* if it is not groupable.

Physical reading. A size is composite iff a heap of that size can crystallize into k identical unit cells; the primes are the non-crystallizable sizes, the stable particles of the decay chains of Remark 4.5. This is, of course, the classical definition of primality. The content is that it is expressible entirely inside the clustered additive world, using only $\odot$ and α, with no imported number labels. One can therefore *define* the particle spectrum as the set of non-groupable heap sizes and generate $\mathcal{T}$ freely on it; the numerical label of a prime leaf is recovered as the observable α of its heap. The foundational question "are prime leaves labeled by known primes, or can the labels be derived?" is thus answered: the labels can be derived. What cannot be derived from the free multiplicative structure alone (§1.4) becomes derivable the moment the additive world and the action are in place, because primality is an observational property of clustering.

Remark 6.8 (Arithmetic functions on trees). Classical arithmetic functions pull back along N: the Euler totient gives $\varphi_{\mathcal{T}}(T) = N(T)\prod_{p\in\mathrm{supp}(T)}(1-1/p) = \varphi(N(T))$, factoring through the double quotient, so it cannot separate $L_2 \boxtimes L_3$ from $L_3 \boxtimes L_2$. A genuinely history-sensitive φ^{tree} would need a recursion in $\boxtimes$ that descends to $\varphi \circ N$ yet distinguishes some equal-shadow pair; likewise a shape-weighted zeta $Z_{\mathcal{T}}(s,q) = \sum_T q^{\mathrm{stat}(T)} N(T)^{-s}$ that escapes P. Both are open (Problem 8.5).

7 Computational content

If the primes are the particles of this gas, can the gas help us find them? Counting cannot, for two reasons given below; the cost of expressions, on the other hand, is where the theory meets the complexity of primes.

First, degeneracy determines primality only circularly: rearranging the recursion of Proposition 3.4,

$$[\, n \text{ prime}\,] \;=\; g_{\mathcal{T}}(n) - \sum_{d\mid n,\, 1<d<n} g_{\mathcal{T}}(d)\, g_{\mathcal{T}}(n/d),$$

but the right side consumes the divisor structure of n, which is what one is trying to find. Second, every degeneracy is *shape-blind*: $g_{\mathcal{X}}(n)$ depends only on the exponent multiset $\{\alpha_p\}$, not on which primes carry them (the closed formulas of Propositions 3.4–3.5 involve only the α_p; for $\mathcal{B}$, the divisor lattice is a product of chains of those lengths). So $12 = 2^2 \cdot 3$, $18 = 2 \cdot 3^2$, and $20 = 2^2 \cdot 5$ share every count ($g_{\mathcal{T}} = 6$, $g_{\mathcal{B}} = 2$, $g_{\mathcal{W}} = 3$), and all counts are computable in $\mathrm{poly}(\log n)$ once n is factored. This is freeness seen computationally: the tree layer is polynomial-time syntax over its input spectrum, so whatever algorithmic leverage exists must come from outside the counting.

What the microstate view does compute is the shape of a typical integer. Draw n uniformly below x and T uniformly from its fiber $N^{-1}(n)$: by the splitting of Proposition 3.4 the shape and leaf-word are independent, the shape uniform Catalan, while Erdős–Kac [15] makes the leaf count $\Omega(n)$ Gaussian around $\log\log x$. So a random integer at full resolution is a uniform random binary tree with $\approx \log\log x$ leaves (five or six for $x = 10^{100}$), carrying no arithmetic information in its shape, of height $\asymp \sqrt{\log\log x}$ [17]. One can even sample a random integer together with its factorization in polynomial time [7, 24]: the gas is efficiently samplable, though deterministically aiming at a prime is not known to be (§7.2).

7.1 Expression cost: the invariant that escapes P

Everything so far in this paper is a functional of the input spectrum: the partition functions of §4 through $P(s), P(2s), \ldots$ (§5), the degeneracies through the exponent profile (shape-blindness above). Here is the first invariant that is not. Extend the grammar of §2 with addition, as anticipated by the bridge: define *expressions*

$$E \;::=\; 1 \;\mid\; (E_1 \oplus E_2) \;\mid\; (E_1 \boxtimes E_2),$$

with evaluator $\mathrm{ev}(1) = 1$, $\mathrm{ev}(E_1 \oplus E_2) = \mathrm{ev}(E_1) + \mathrm{ev}(E_2)$, $\mathrm{ev}(E_1 \boxtimes E_2) = \mathrm{ev}(E_1)\,\mathrm{ev}(E_2)$, and cost the number of 1-leaves. The minimum cost

$$\|n\| \;=\; \min\{\, \mathrm{cost}(E) : \mathrm{ev}(E) = n \,\}$$

is the classical *integer complexity* of Mahler–Popken [29] (OEIS A005245 [35]), with $3\log_3 n \le \|n\| \le 3\log_2 n$ for $n > 1$; allowing subexpression reuse (straight-line programs) gives the Shub–Smale measure $\tau(n) = O(\log n)$ [34].

Cost sees what no degeneracy can: $\|4\| = 4$ while $\|9\| = 6$, although $4 = 2^2$ and $9 = 3^2$ have the same exponent profile. Structurally it is a bridge quantity: it measures how expensively an integer can be synthesized once additions are allowed inside the build, and it is where the known connections between the structure of multiplication and the complexity of primes live:

Proposition 7.1 (Wilson's theorem as an expression statement). *(i) $n \ge 2$ is prime if and only if $(n-1)! \equiv -1 \pmod{n}$.*

(ii) If $(x, N) \mapsto x! \bmod N$ were computable in time polynomial in $\log N$, then factoring would be in polynomial time.

Proof. (i) is Wilson's theorem. (ii): $\gcd(x! \bmod N,\, N) = \gcd(x!,\, N)$, which equals 1 exactly when x is smaller than the least prime factor of N and exceeds 1 from that point on; binary search over $x \in [1, N]$ therefore finds the least prime factor with $O(\log N)$ evaluations, and recursion factors N completely. □

The same object carries the deeper conjectures. If $n!$ had straight-line programs of size polylogarithmic in n, factoring would have polynomial-size circuits (folklore; see Blum–Cucker–Shub–Smale [9]), and the Shub–Smale τ-conjecture (a polynomial bound on the integer roots of a polynomial in terms of its program size) implies $\mathrm{P}_{\mathbb{C}} \neq \mathrm{NP}_{\mathbb{C}}$ [34]. In the vocabulary of this paper: the computational hardness of arithmetic lives at the additive–multiplicative interface of §6, in the cost of synthesizing specific integers with additions, and Wilson's theorem puts primality one factorial-expression away.

The theory also suggests its own cost observable. Define the *factored cost* and the *addition dividend*

$$A(n) \;=\; \sum_{p^a \| n} a\,\|p\|, \qquad \delta(n) \;=\; A(n) - \|n\| \;\ge\; 0:$$

$A(n)$ is the cost of the cheapest expression that respects the prime factorization (synthesize each prime additively, then multiply), and $\delta(n)$ is the saving that additions *above* the factorization can buy. The long-standing open question whether $\|2^k\| = 2k$ is exactly the statement $\delta \equiv 0$ on powers of two; and δ is not identically zero: $\delta(55) \ge 1$, since $A(55) = \|5\| + \|11\| = 13$ while $55 = 2 \cdot 27 + 1$ gives $\|55\| \le 12$. Nothing about the distribution of δ appears to be known; see Problem 8.7.

7.2 What finding a prime costs

Recognizing primes is solved: PRIMES $\in$ P [1]. Hitting one randomly is easy: by the prime number theorem a random n-bit integer is prime with probability $\sim 1/(n \log 2)$, so sampling plus testing generates an n-bit prime in expected time $\widetilde{O}(n^3)$ [14]. The genuine open problem, the Polymath 4 problem [33], is deterministic generation:

task	best known	status
test an n-bit integer	deterministic poly(n) [1]	solved
generate, randomized	$\widetilde{O}(n^3)$: sample and test	routine
generate, deterministic	$\sim 2^{n/2}$: analytic $\pi(x)$ methods [33]	open, even under RH
generate, pseudodeterministic	poly(n), infinitely many n [12]	frontier
quantum search	$\sim 2^{0.2625\,n}$: Grover [20] on [8]	quadratic-type gain

Even the Riemann Hypothesis only bounds prime gaps by $O(\sqrt{x}\log x)$, leaving a $2^{n/2}$ search; a Cramér-type gap bound $O(\log^2 x)$ [13] would make "scan forward and test" polynomial. The two available attack surfaces, gap bounds and derandomization, are the two places where this paper has located the remaining hardness. Gap bounds are quantitative statements about the additive–multiplicative interface: §6 names the interface but is not yet quantitative. Derandomization is a problem of aiming: the primes are dense and efficiently recognizable, and a deterministic generator is, in the end, a short expression that evaluates to a prime, which places the problem in the expression-cost frame of §7.1. In summary, the theory maps where the hardness cannot be (shape-blind counting) and poses cost questions at the interface where it can be; it does not provide a shortcut.

8 Open problems

Question 8.1 (Tree-zero distribution: beyond the almost periodic region)**.** Theorem 5.3 settles the density of tree zeros in substrips of $\operatorname{Re} s > 1$, and Corollary 5.7 evaluates the total density $D(1) = 0.1436\ldots$ Remaining: (a) the left edge: the census finds real parts down to 1.26 within the scanned box and lower on coarse sweeps; do the zeros accumulate at 1? Do zeros of the continued $P - \frac{1}{4}$ exist with $\operatorname{Re} s \le 1$, where almost periodicity fails and the Riemann zeros enter through (5.1)? (b) upgrade $D(1)$ to a boundary count: is the number of zeros with $\operatorname{Re} s > 1$ and $0 < t \le T$ asymptotic to $D(1)\,T$, or does accumulation at the line $\operatorname{Re} s = 1$ add a correction? (c) prove the observed rigidity: the empirical number variance saturates ($\Sigma^2(10) = 0.67$ against 10 for Poisson) and the spacings repel (Figure 5); establish that the tree zeros form a quasi-periodic set with a controlled defect density, and compute the pair-correlation function. (d) a Lee–Yang-type question: is there a natural deformation of the gas for which the zeros align on a line? (e) the same program for the level sets $P(s) = c$, $0 < c < 1$, of which the tree zeros are $c = \frac{1}{4}$ and the Boltzmann-gas poles $c = 1$: the fugacity of Remark 4.6 realizes the family $c = 1/(4z)$.

Question 8.2 (Tauberian rigor)**.** Prove $\sum_{n\le x} g_{\mathcal{T}}(n) \sim c\,x^{\sigma^*}(\log x)^{-3/2}$ (a Delange-type theorem for Dirichlet series with a square-root singularity), and its analogue for $g_{\mathcal{B}}$.

Question 8.3 (The cascade of the non-planar gas)**.** The continuation

$$Z_{\mathcal{B}}(s) = 1 - \sqrt{1 - 2P(s) - Z_{\mathcal{B}}(2s)}$$

propagates each singularity of $Z_{\mathcal{B}}$ at s_0 to potential singularities at $s_0/2, s_0/4, \ldots$ and mixes them with those of P. Map this self-similar singularity structure; determine whether $Z_{\mathcal{B}}$ has a natural boundary strictly to the right of $\operatorname{Re} s = 0$.

Question 8.4 (Beyond the single-particle input)**.** Every one-variable partition function of §4 is a functional of $P(s), P(2s), \ldots$ Find a natural two-variable $Z_{\mathcal{T}}(s, q)$ (weighting a shape statistic of the history) that is provably *not* such a functional, and study its phase diagram in (s, q).

Question 8.5 (History-sensitive totient)**.** Construct φ^{tree} as in Remark 6.8 (a $\boxtimes$-recursion descending to $\varphi \circ N$ that separates some equal-shadow pair), ideally with an inclusion–exclusion interpretation over subtrees.

Question 8.6 (The full expression calculus)**.** §6 lets processes act on states, and §7.1 equips the two-sorted expression grammar with a cost. What is missing is its *rewriting theory*: treat distributivity as a directed rewrite rule (an evaluation step) rather than an equation, study normal forms and confluence, and relate them to Loday's grove addition [28, 6].

Question 8.7 (The addition dividend)**.** Determine the behavior of $\delta(n) = A(n) - \|n\|$ (§7.1). Is $\delta \equiv 0$ on powers of two (equivalently, $\|2^k\| = 2k$)? Is δ unbounded? What is its average order? Is $\|n\|$ computable in time polynomial in $\log n$? And the structured refinement: quantify the *cost of remembering*, the minimal cost of an expression whose multiplicative skeleton is constrained to a fixed corner of the coarse-graining square.

Question 8.8 (Axiomatics of the bridge)**.** Characterize the pair $(\mathcal{G}, \alpha)$ abstractly: for which "clustering worlds" does Theorem 6.4 hold with exactly $\approx_a$, and is $(\mathcal{G}, \alpha)$ universal among them?

Acknowledgments

During the preparation of this work the author used an AI system (Claude, Anthropic) for drafting, computation, and verification. The author reviewed, edited, and verified all content and takes full responsibility for it. All quantitative claims are independently reproducible from the accompanying code (see the data availability statement).

Data and code availability

All computations in this paper are reproducible from the open-source `primeleaf` Python package and scripts at `https://github.com/gbarbalinardo/primeleaf`, archived on Zenodo at `https://doi.org/10.5281/zenodo.21252898`. The repository includes the computed tree zeros to height 5000, the density curve, the figure-generation scripts, and a test suite; the commands reproducing each part of Appendix A are listed there.

A Computational verification

All computations are reproducible from the accompanying `primeleaf` Python package (see the repository `README` for installation): `python -m primeleaf verify-combinatorics` reproduces the combinatorial checks below, `verify-zeta` the identities and constants, and `census` and `dominance` the zero census and the density check.

A.1 Combinatorial checks

For every $n \le 20000$: the closed formula of Proposition 3.4 agrees with a direct recursive enumeration $g_{\mathcal{T}}(n) = [n \in \mathbb{P}] + \sum_{d|n,\, 1<d<n} g_{\mathcal{T}}(d)\, g_{\mathcal{T}}(n/d)$; the multinomial of Proposition 3.5 agrees with the first-letter recursion; and the unordered recursion of Proposition 3.6 reproduces the Wedderburn–Etherington numbers $1, 1, 1, 2, 3, 6, 11, 23, 46, 98, 207, 451, 983, 2179$ at $n = 2^k$, $k \le 14$, as well as the hand-enumerated values $g_{\mathcal{B}}(12) = 2$, $g_{\mathcal{B}}(24) = 4$, $g_{\mathcal{B}}(30) = 3$. The original

formula and program of OEIS entry A375120 used ordered pairs on the diagonal and first diverged from the unordered count at $n = 144$ (42 against the true 41); following this comparison the entry was corrected in July 2026 and now lists the unordered counts to $n = 10000$ (Remark 3.7).

A.2 Analytic checks and constants

With $g_{\mathcal{T}}(n)$, $g_{\mathcal{B}}(n)$ enumerated for $n \leq 20000$, partial sums of the Dirichlet series were compared with the closed/functional forms of Theorem 4.1:

series	s	partial sum vs. closed form	difference
$Z_{\mathcal{T}}$	3	$0.225547637865\ldots$ vs. $0.225705704214\ldots$	$-1.6 \cdot 10^{-4}$ (tail)
$Z_{\mathcal{T}}$	4	$0.084059063652\ldots$ vs. $0.084059066410\ldots$	$-2.8 \cdot 10^{-9}$
$Z_{\mathcal{T}}$	6	agreement	$-3.1 \cdot 10^{-18}$
$Z_{\mathcal{B}}$	3	agreement	$-4.0 \cdot 10^{-7}$
$Z_{\mathcal{B}}$	4	agreement	$-1.1 \cdot 10^{-11}$

the residual differences being exactly the truncation tails expected from the critical exponents. Constants (via `mpmath.primezeta` at 25–30 significant digits):

$$\begin{aligned} \sigma^* &= 2.59738512713467162\ldots, & P'(\sigma^*) &= -0.230054805474\ldots, \\ \sigma_{\mathcal{B}} &= 1.98847685180090810\ldots, & Z_{\mathcal{B}}(\sigma_{\mathcal{B}}) &= 1 \text{ (verified to } 10^{-6}), \\ \sigma_{\mathcal{W}} &= 1.39943332872633032\ldots, & P'(\sigma_{\mathcal{W}}) &= -1.73115620186\ldots, \\ \rho_{\text{Kalmár}} &= 1.72864723899818362\ldots, & P(2) &= 0.45224742004106550\ldots \end{aligned}$$

For the classical bootstrap of Remark 4.5 on the prime spectrum: Boltzmann counting, $P(\beta_H) = 2\log 2 - 1$, gives $\beta_H = 2.14875116590273\ldots$; Bose counting, solving $P(s) + 2\sum_{k\geq 2} G(ks)/k = 2\log 2 - 1$ with G computed by the argument-doubling cascade, gives $\beta_H = 2.30723145510135\ldots$ with $G(\beta_H) = 0.6646652572\ldots$

A.3 Tree zeros

The zeros of Table 3 were located by a grid scan of $|P_{\leq 20000}(s) - \frac{1}{4}|$ (truncated prime sum, step 0.02) over $[1.40, 2.70] \times [0.3, 50]$, followed by Newton polishing on the full `primezeta` at 20 significant digits; each reported zero satisfies $|P(s) - \frac{1}{4}| < 10^{-15}$. A coarser sweep of $[1.05, 1.40) \times [0.5, 50]$ (step 0.05×0.25) found no additional candidates; its only sub-threshold dip flowed to zero #1 of the table.

A.4 Zero census, density curve, and spacings

Census (§5.3, §5.4, command `python -m primeleaf census`). The scan covers $[1.25, 2.72] \times [0.3, 5000]$ with primes up to 2×10^5 (grid 0.02×0.05, seed threshold 0.09, Newton polish on the full `primezeta` at 18 significant digits, residuals below 10^{-14}): 595 zeros, real parts in (1.2615, 2.5679), all below σ^*, listed in `data/zeros_t5000.txt`. The mean spacing is 8.39 against $2\pi/\log 2 = 9.06$. The figures are produced by `scripts/make_figures.py`.

Density curve (command `python -m primeleaf density`). Dominance probabilities from 10^6 random-phase samples over the primes up to 2×10^4, for the first 200 primes, with an analytic tail $\frac{1}{2\pi}\sum_{p>p_0}(\log p)\,\pi p^{-2\sigma} f_{\text{loc}}$ (the tail never exceeds 0.001):

σ	1.30	1.20	1.10	1.05	1.02	1.01	1.00
$D(\sigma)$	0.1162	0.1223	0.1314	0.1372	0.1406	0.1421	0.1435

with, at $\sigma = 1.30$, $q_2 = 0.624$, $q_3 = 0.148$, $q_5 = 0.042$, $q_7 = 0.017$. The total density $D(1)$ was pinned by four independent seeds (1.5×10^6 samples each): $0.14363, 0.14341, 0.14367, 0.14352$, mean 0.14356, standard deviation 0.00010; we quote $D(1) = 0.1436$. As a cross-check at $\sigma = 1.30$, $D(1.30) = 0.1162$ predicts 58.1 zeros to height 500, against the 59 found by the earlier dedicated $t \leq 500$ census.

Spacings (§5.4). From the 595 ordinates, normalized to unit mean: coefficient of variation of the raw gaps 0.259; smallest normalized gap 0.79; fraction below $s = \frac{1}{2}$ equal to 0.081 (Poisson: 0.393); number variance $\Sigma^2(1) = 0.18$, $\Sigma^2(5) = 0.57$, $\Sigma^2(10) = 0.67$ (Poisson: $\Sigma^2(L) = L$). The saturation of Σ^2 and the near-total level repulsion are the rigidity reported in the text.

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